

Lab 2 — Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement #26: Peer-to-Peer Carpooling & Expense Splitter

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1. Epics Created

Three epics were created to group the functional requirements:

Epic 1: Route Matching & Ride Discovery — Enable riders and drivers to register commute routes and find compatible ride matches within detour tolerance.

Epic 2: Fuel Cost Splitting & Payments — Calculate and process automated fuel cost share splits between driver and passengers.

Epic 3: Trust & Safety — Allow users to rate co-travelers and store sensitive documents securely.

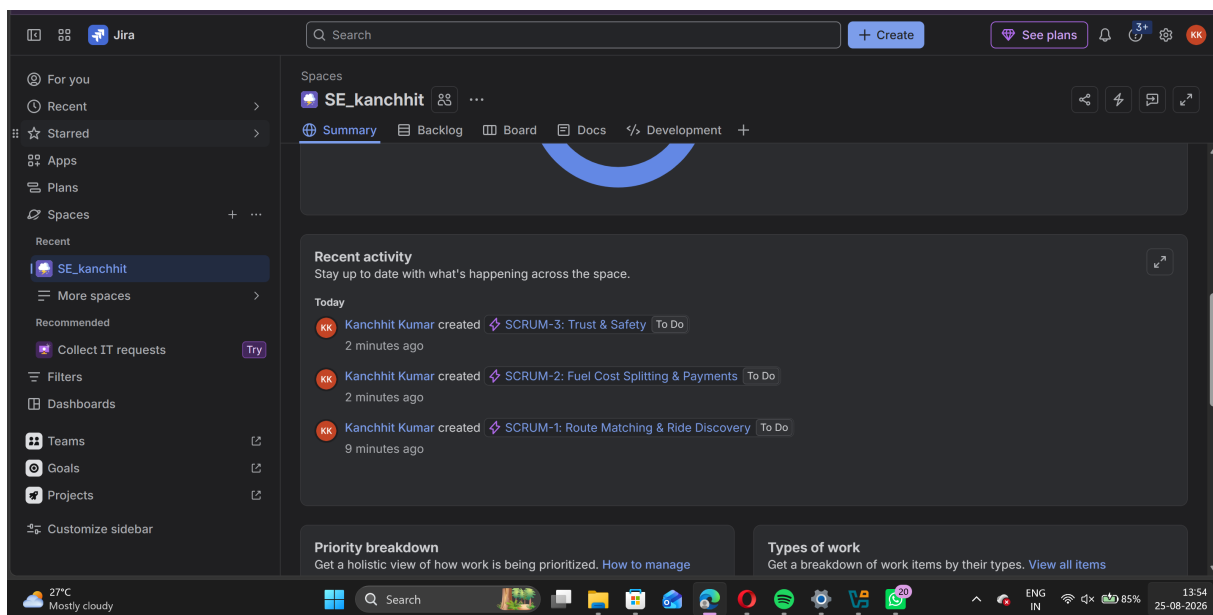


Fig 1: All three Epics created in Jira (Summary view)

2. Backlog with User Stories & Story Points

Eight user stories were created under their respective Epics, with story points assigned using the Fibonacci scale:

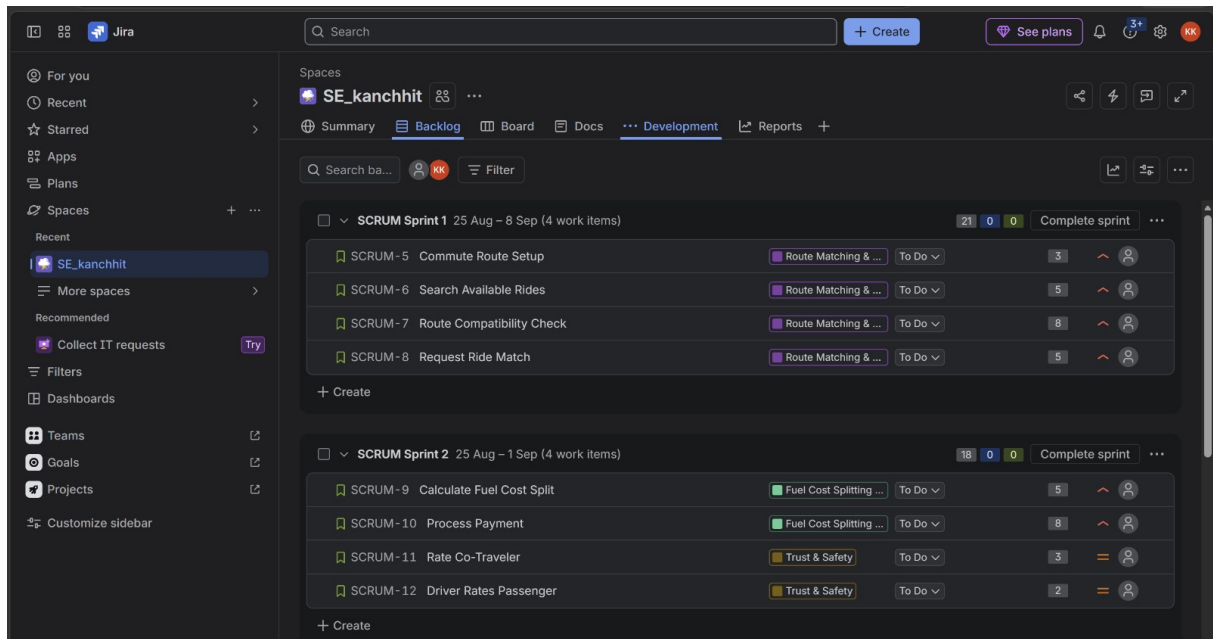


Fig 2: Backlog showing Sprint 1 (21 pts) and Sprint 2 (18 pts) with story point assignments

3. Sprint Board (Active Sprint View)

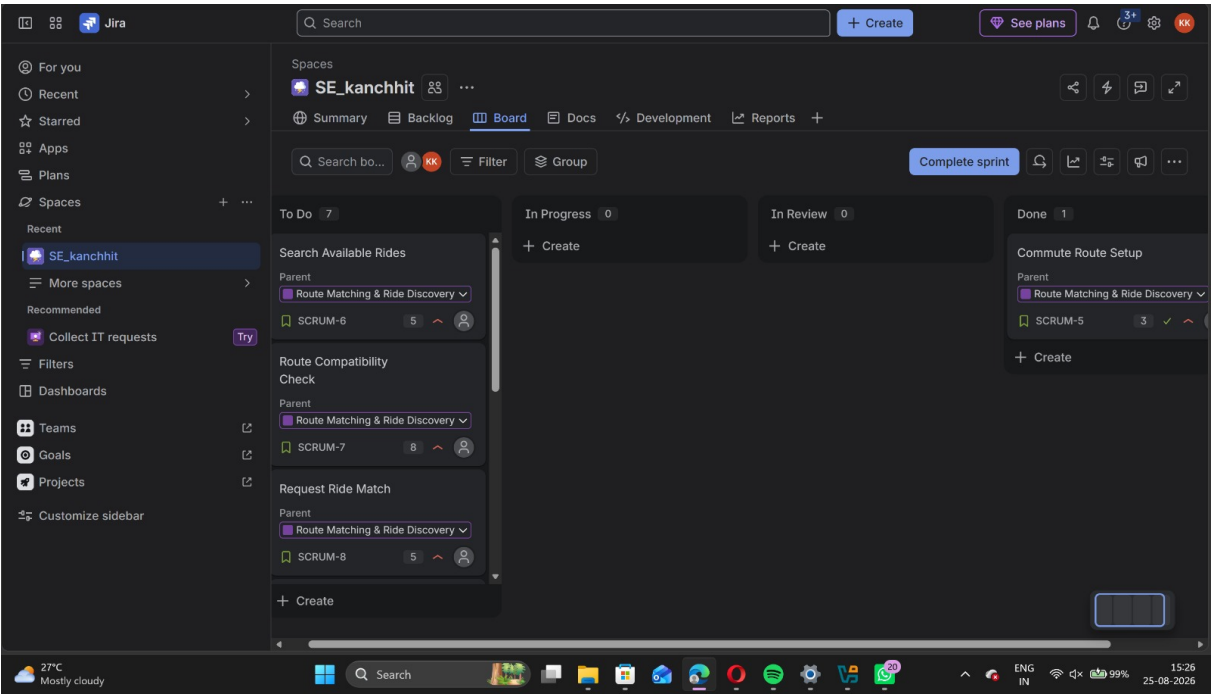


Fig 3: Sprint 1 board — stories moving from To Do to Done

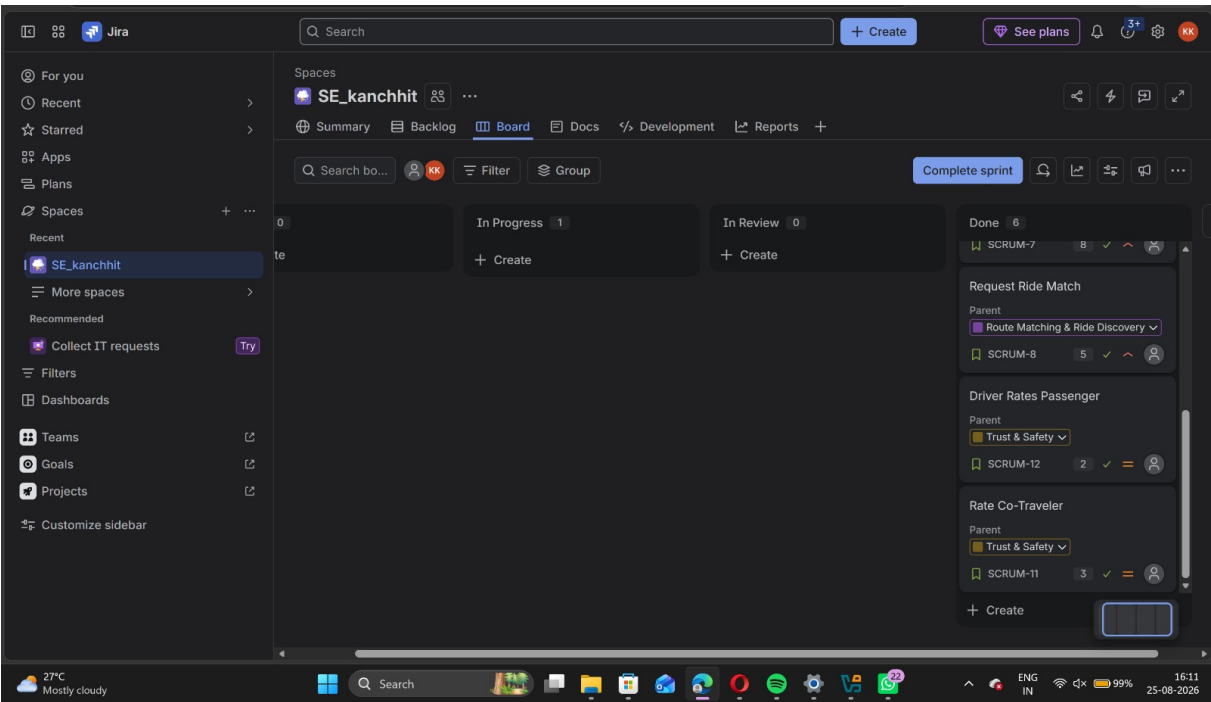


Fig 4: Sprint 2 board — all stories completed in Done column

4. Burndown Charts

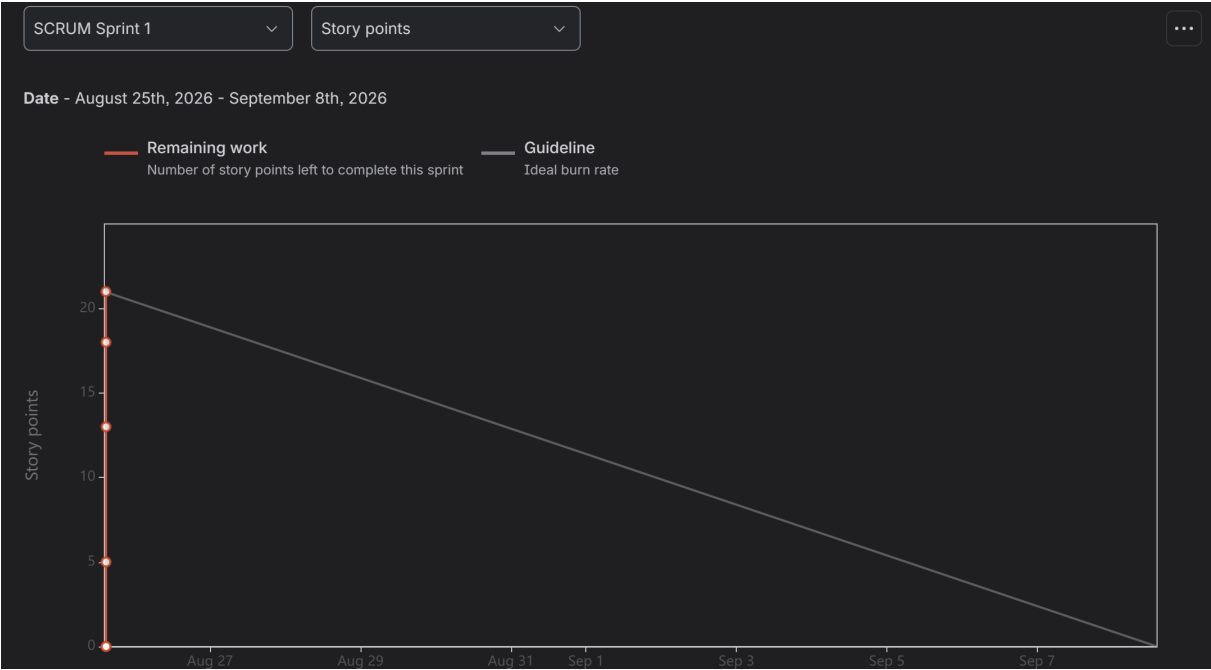


Fig 5: Sprint 1 Burndown Chart — 21 story points completed

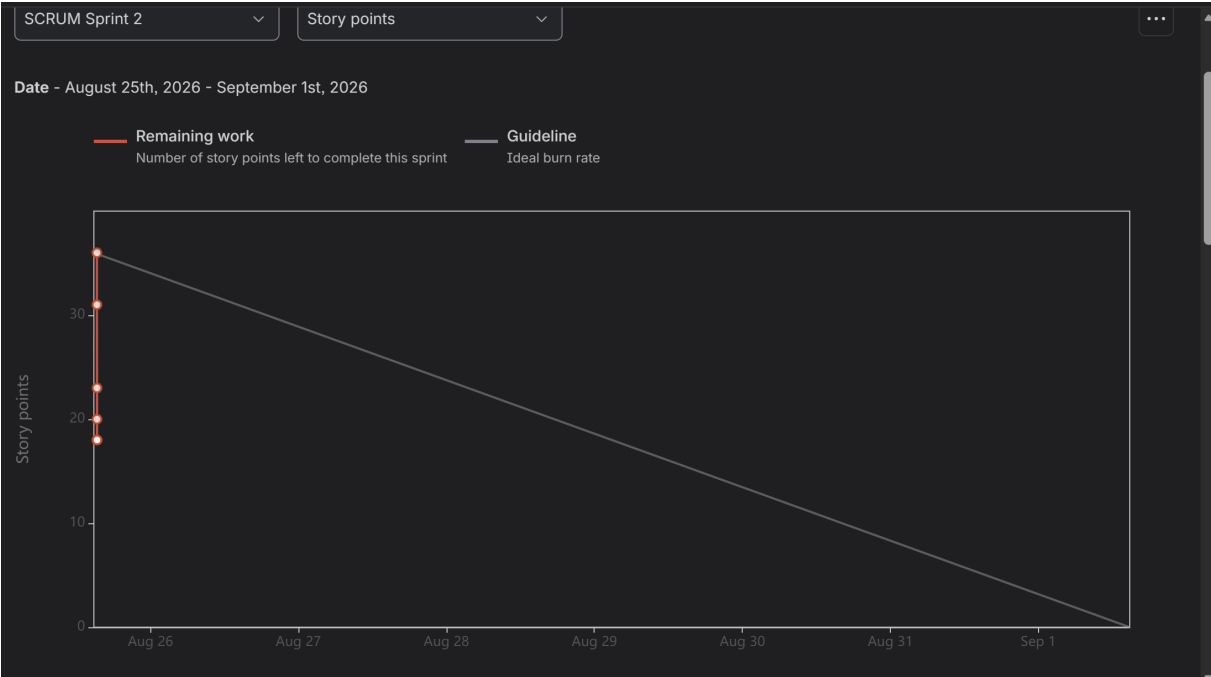


Fig 6: Sprint 2 Burndown Chart — 18 story points completed

5. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The higher-point stories (Route Compatibility Check and Process Payment at 8 pts each) were correctly identified as the most complex since they involve geospatial calculations and payment gateway integration respectively. Lower-point stories like Driver Rates Passenger (2 pts) were straightforward UI work.

2. Was your backlog well-prioritized?

Yes. Core route matching features were placed in Sprint 1 since no other functionality works without them — riders can't split costs or rate drivers if they can't find a ride first. Payment and trust features were in Sprint 2 as they depend on the matching system being in place.

3. How did your simulated sprint align with your plan?

Both sprints completed on schedule with all stories moved to Done within the sprint window. Sprint 1 had 21 story points and Sprint 2 had 18 points, both manageable within a 1-week duration. In a real project, the velocity from Sprint 1 (21 pts) would help plan Sprint 2's capacity.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts show a steep early drop because stories were completed quickly in the simulation. In a real scenario, a more gradual decline following the guideline would indicate healthy pacing. The sharp drop suggests that in practice, either the team has excess capacity (could take on more points) or the work wasn't spread evenly across the sprint. The guideline helps identify whether the team is ahead or behind schedule.